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Scalable biochar production from oil palm fronds and waste cooking oil: Techno-economic and policy-oriented evaluation for rural deployment

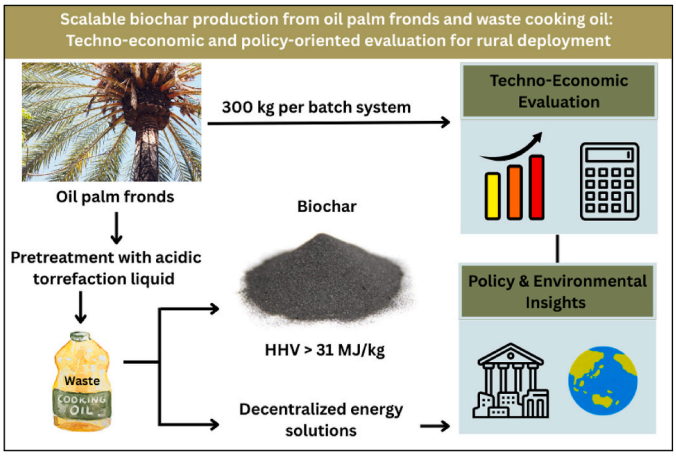
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HIGHLIGHTS

- Oil palm fronds were pretreated using acidic torrefaction water before co-torrefaction.
- Co-torrefaction with waste oil improved biochar higher heating value to over 31 MJ/kg.
- 300 kg batch system demonstrated strong techno-economic feasibility for rural use.
- Feedstock cost and market price are key variables in the sensitivity analysis outcomes.
- Findings offer policy and environmental insights for small-scale biomass valorization.

GRAPHICAL ABSTRACT



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ABSTRACT

The increasing demand for sustainable biomass utilization and decentralized energy solutions has highlighted the need for low-cost biochar systems in rural areas. This study assesses the techno-economic feasibility of producing premium biochar from oil palm fronds (OPF) and waste cooking oil (WCO) through acid pretreatment and co-torrefaction in a 300-kg batch system, suitable for community-scale deployment. The process achieved a biochar yield of 60.95 %, an energy yield of 87.43 %, and a high heating value of 31.97 MJ/kg, classifying the product as premium-grade. Financial modelling included capital and operational costs, break-even point (BEP), net present value (NPV), and internal rate of return (IRR), supported by sensitivity analysis under varying market

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conditions. Under moderate scenarios, the system achieved an IRR exceeding 40 % and a payback period of less than two years. The analysis further explored scaling strategies, carbon offset potential, and enabling policy mechanisms relevant to palm oil-producing regions. Each batch was estimated to sequester 0.55 t-CO_2-eq, aligning with circular bioeconomy goals and rural climate mitigation strategies. The model demonstrates that integrating biochemical pretreatment with co-torrefaction offers a viable and replicable solution for sustainable waste valorization, energy access, and carbon management in Southeast Asia. This study addresses key knowledge gaps by linking process optimization with economic and policy analysis, offering practical insights for cleaner production in decentralized settings.

1. Introduction

The global shift toward sustainable energy and waste reduction has renewed interest in converting agricultural residues into renewable fuels (Dewi et al., 2024). In Indonesia, the world's largest palm oil producer (Purnama et al., 2025), over 250 million tons (Mt) of oil palm fronds (OPF) are generated annually, most of which remain unmanaged and contribute to air pollution and GHG emissions (Ooi et al., 2017; Purnama et al., 2024a).

OPF possesses favorable properties for biochar production, including high volatile matter and fixed carbon, low ash and sulphur, and a fibrous structure that supports porosity and char stability (Jadhav et al., 2021; Nabila et al., 2023). However, its high alkali content, particularly potassium, necessitates pretreatment to enhance product quality (Tang et al., 2022). Several pretreatment strategies, including acid leaching and co-processing with lipids, have enhanced biomass conversion (Pagano et al., 2023; Yek et al., 2021). However, limited studies have applied these approaches to OPF or assessed their techno-economic potential in rural tropical settings. Mutamima et al. (2025) demonstrated that combining acid pretreatment and co-torrefaction with waste cooking oil (WCO) improves biochar fuel quality, increasing the higher heating value (HHV) to 31.97 MJ/kg, the mass yield to 60.95 %, and the energy yield to 87.43 %.

Despite promising technical outcomes, the economic viability of biochar production systems remains underexplored—particularly in tropical regions where agricultural waste is abundant but capital access is limited. While most studies focus on thermochemical performance or environmental benefits, techno-economic evaluations, which are essential for real-world deployment, are often overlooked (Abhijeet et al., 2020; Layek et al., 2022). A few recent efforts have estimated production costs for biochar from sugarcane bagasse, forest residues, or oil palm biomass (Parthasarathy et al., 2024; Selvam et al., 2024); however, none have comprehensively assessed OPF-based systems that integrate pretreatment and co-torrefaction.

Using waste-derived oil as a co-feedstock introduces logistical and pricing complexities, requiring a sensitivity analysis to evaluate financial robustness. However, the growing availability of WCO from urban food sectors supports its inclusion in biomass conversion and aligns with the principles of the circular bioeconomy (Ulloa-Murillo et al., 2022). Still, technical efficiency alone is insufficient; long-term viability depends on economic resilience and scalability in varied rural contexts.

This study adopts a cleaner production and circular bioeconomy framework, integrating acid pretreatment and lipid-assisted co-torrefaction as a hybrid upgrading strategy. Informed by decentralized energy justice principles, the proposed small-scale model promotes equitable energy access and local value creation. It aims to investigate whether a small-scale co-torrefaction system utilizing acid-pretreated OPF and WCO can simultaneously achieve premium biochar quality, economic feasibility, and policy relevance in rural Indonesia. Accordingly, this work provides a techno-economic assessment of biochar production from acid-pretreated OPF and WCO, analyzing energy yields, cost structure, financial indicators, and sensitivity scenarios in Riau Province, Indonesia. It bridges the gap between technical optimization and policy-aligned, scalable deployment for biomass-rich rural settings.

2. Literature review

2.1. Biochar production technologies

Biochar is increasingly valued for its role in improving soil, sequestering carbon, and managing waste. Thermochemical methods—pyrolysis, gasification, and torrefaction—are widely used, each offering trade-offs in energy efficiency and product quality. Torrefaction, a low-temperature process (200–300 °C), enhances the properties of biomass fuels by reducing moisture and volatiles while increasing energy density (Purnama et al., 2024b). Recent innovations such as co-torrefaction and acid pretreatment using torrefaction condensates further improve reactivity and char stability (Chen et al., 2017).

As shown in Fig. 1, combining acid pretreatment with co-torrefaction using WCO can yield premium biochar with higher carbon content and heating value. This approach remains underexplored for tropical biomass, such as OPF, despite its abundance and favorable lignocellulosic composition.

2.2. Economic analysis of bioenergy systems

The viability of bioenergy systems depends not only on technical performance but also on cost-effectiveness, particularly in decentralized contexts. Numerous techno-economic studies have assessed pyrolysis and biochar production in rural settings using key indicators such as net present value (NPV), internal rate of return (IRR), and payback period (PP) (Parthasarathy et al., 2024). As shown in Table 1, most models focus on large-scale, centralized systems with continuous processing and single feedstocks. While hybrid systems—e.g., biochar slurry fuels or catalytic additives—offer improved profitability, they often entail higher capital costs. Decentralized, batch-scale models remain underrepresented in the literature. This study addresses that gap by evaluating a small-scale co-torrefaction system using acid-pretreated OPF and WCO, demonstrating strong IRR and short PPs even at modest capacity.

2.3. Agrowaste co-valorization and enabling policies

Blending agricultural biomass with oily residues, such as WCO, through co-processing enhances char quality and energy yield by improving combustion efficiency and calorific value (Lam et al., 2019; Mutamima et al., 2025). This approach supports circular economy goals by diversifying waste utilization pathways. OPF, although abundantly available (exceeding 200 Mt/y in Southeast Asia), remains underutilized. Their fibrous structure, low ash content, and moderate lignin content make them ideal for conversion, while synergy with WCO links rural biomass with urban waste streams, creating a rural–urban valorization model.

For successful deployment, policy support is crucial. Decentralized biomass systems often face regulatory and financial barriers. In Indonesia, programmes such as subsidized electricity tariffs and *Kredit Usaha Rakyat* (KUR) loans offer potential support for rural enterprises. Moreover, institutional anchors like BUMDes and farmer cooperatives can serve as local implementers. Embedding biochar initiatives within these structures—aligned with national goals for waste reduction and

low-carbon development—can facilitate scalable, community-driven adoption (Ulloa-Murillo et al., 2022).

2.4. Research gap and originality of this study

While numerous studies have examined pyrolysis and torrefaction individually, few have investigated the integrated approach of acidic pretreatment followed by co-torrefaction, especially in community-scale systems. Most existing techno-economic models either focus on centralized facilities or omit policy and institutional contexts relevant to rural Southeast Asia. Moreover, there is limited empirical data on the production of premium-grade biochar using OPF and WCO, despite their abundance and complementary characteristics.

This study bridges these gaps by combining experimental parameters from earlier trials (Mutamima et al., 2025) with a detailed techno-economic model tailored for decentralized rural contexts. It also incorporates policy scenario analysis, offering a holistic framework for scalable biomass valorization, particularly in palm oil-producing regions of Southeast Asia where such models remain scarce.

3. Methodology

3.1. Scope and data source

This techno-economic assessment does not involve new laboratory work but draws on empirical findings from Mutamima et al. (2025), which optimized a co-torrefaction process using acid-pretreated OPF and WCO. The current study evaluates its feasibility at the community scale using a 300-kg batch model—typical of decentralized biochar units operated by farmer cooperatives, which process around 1200–2000 kg of biomass weekly. Key technical data adopted from Mutamima et al. (2025) include a biochar yield of 60.95 %, an energy yield of 87.43 %, and an HHV of 31.97 MJ/kg. These were used as input parameters for the mass and energy balance calculations, as well as for financial simulations under multiple operational scenarios (Parthasarathy et al., 2024; Selvam et al., 2024). In Mutamima et al. (2025), biochar characterization included yield measurement, determination of calorific value via bomb calorimetry (ASTM D5865), elemental analysis (CHN), and proximate analysis using thermogravimetric analysis (TGA, LECO TGA701). In addition, structural properties were assessed using Fourier transform infrared spectroscopy (FTIR), X-ray diffraction (XRD), and TGA.

To obtain site-specific cost data, field visits and interviews were conducted in Riau and West Sumatra with equipment manufacturers and biomass operators during late 2024. Information on fabrication, equipment prices, and labor was collected directly from workshops producing torrefaction units. OPF and WCO prices were sourced through market surveys and consultations with cooperatives and vendors, while electricity and wage data were validated using official statistics and input from stakeholders. All data were triangulated and applied in Sections 3.2–3.5.

3.2. Production system and cost modeling

The production model utilizes 300 kg of dried OPF per batch, reflecting small-scale systems suitable for farmer cooperatives. The process involves five stages: (1) chopping and sun-drying OPF; (2) acid pretreatment by soaking in torrefaction liquid (pH 3, 50 °C, 75 min), followed by rinsing and drying; (3) blending with WCO; (4) co-torrefaction at 250 °C in a low-oxygen stainless steel reactor; and (5) cooling, drying, and packaging. As shown in Fig. 2, these steps replicate the optimized conditions from Mutamima et al. (2025) and underpin the cost estimation in this study. The resulting biochar contains >70 % fixed carbon, <5 % ash, and particle size <2 mm—properties consistent with premium-grade standards for soil and carbon applications (Jadhav et al., 2021; Mutamima et al., 2025).

3.2.1. Capital investment estimation

The capital investment estimation utilizes a structured cost breakdown to enhance replicability and accuracy, surpassing the limitations of informal quotations. The breakdown covers equipment procurement, installation, civil works, instrumentation, engineering overhead, contingency, and initial working capital. A summary is provided in Table 2.

While the exact application of Lang factor estimation is limited by the small scale and rural-specific variability, the combined capital structure reflects an approximate Lang factor of 3.2, which aligns with semi-automated, decentralized biomass systems (Turton et al., 2018). This structured estimation approach enhances reproducibility and facilitates integration into process simulation platforms such as CapCost or Aspen Process Economic Analyzer for scaling across diverse contexts. Equipment depreciation was calculated using the straight-line method, assuming a lifespan of five years for the stainless-steel torrefaction reactor and three years for auxiliary equipment, which is typical for community-scale setups. These depreciation values were incorporated into the annual cash flow projections.

3.2.2. Operational cost estimation

Operational costs for a single 300-kg batch include feedstock procurement, energy consumption, labor, and process consumables. OPF, abundantly available in Riau—the largest oil palm-producing province in Indonesia—is considered a low-cost agricultural residue. A nominal collection and transportation cost of IDR 800/kg was applied, reflecting informal labor rates based on community harvesting schemes. WCO, sourced from small food vendors, is priced at IDR 3500/kg based on local market surveys conducted in Pekanbaru, Riau, Indonesia.

Electricity use (68 kWh per batch) covers drying, heating for pretreatment (50 °C for 75 min), and torrefaction. Labor cost is estimated at IDR 300,000 per batch, equivalent to the prevailing rural daily wage in Riau for one operator handling the entire process. Consumables—including acid, water, and personal protective equipment—are assigned a lump-sum cost of IDR 90,000, derived from actual quantities used in pilot operations. The full operational cost breakdown is presented in Table 3.

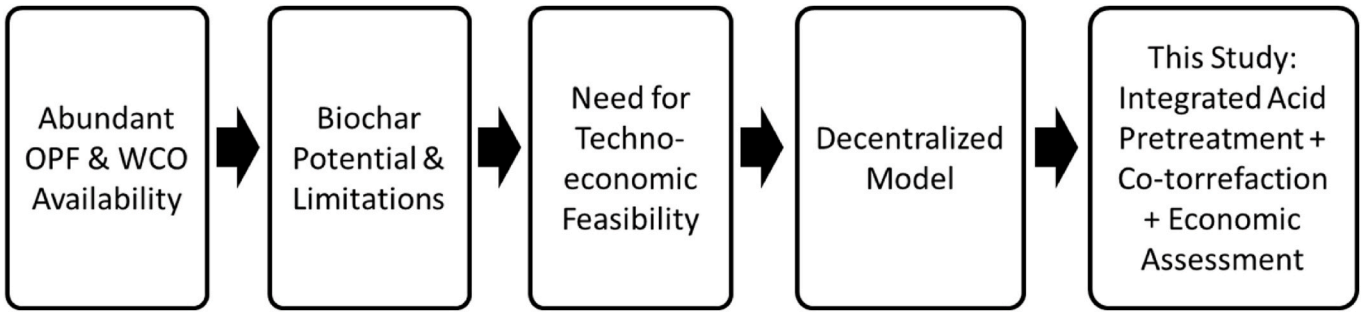


Fig. 1. Conceptual framework illustrating the integration of acid pretreatment and co-torrefaction of OPF with WCO for decentralized premium biochar production and carbon offset.

With a biochar yield of approximately 60.95 %, each batch produces approximately 183 kg of solid product. This results in a unit production cost of IDR 9738 per kg, or around USD 0.62/kg. The modeling approach mirrors decentralized energy systems and reflects context-appropriate economic assumptions for community-scale biochar initiatives in Indonesia (Owsianiak et al., 2021; Sahoo et al., 2021).

3.3. Financial metrics and scenario analysis

To evaluate the financial viability of the proposed small-scale biochar system, we calculated four key indicators: break-even point (BEP), NPV, IRR, and PP. These metrics were assessed under three pricing scenarios (conservative, moderate, and optimistic) and two operational intensities (2 and 3 batches per week).

The modeling approach was adopted based on standard practices in process economics for small-scale biomass valorization (Peters et al., 2021; Turton et al., 2018; Hu et al., 2024). The BEP, expressed as the number of batches required to recover the initial investment (C_0), was calculated using:

$$BEP = \frac{C_0}{\pi} \tag{1}$$

where:

C_0 = initial capital investment (IDR 150,000,000),
π = net profit per batch under each scenario.

The NPV over a 5-year project horizon was estimated using a discount rate (r) of 10 %:

$$NPV = \sum_{t=1}^{5} \frac{P}{(1+r)^t} - C_0 \tag{2}$$

where:

P = annual profit (P = π × n),
n = number of batches per year (assuming 2–3 batches/week for 48 weeks),
r = 0.10.

The IRR was calculated by numerically solving for the discount rate at which NPV equals zero (Peters et al., 2021). The PP, representing the time required to recover the investment, was calculated as:

$$PP = \frac{C_0}{\text{Annual Net Profit}} \tag{3}$$

Three market price scenarios were considered for premium biochar: IDR 12,500/kg (conservative), 15,000/kg (moderate), and 17,500/kg (optimistic), reflecting actual retail prices in the horticulture, carbon sequestration, and high-end soil amendment markets (Mutamima et al., 2025). All calculations were implemented using Microsoft Excel 2019 (Microsoft Corporation, Redmond, WA, USA) with iterative solver tools to compute IRR and simulate sensitivity scenarios.

3.4. Sensitivity and risk analysis

To assess the economic robustness of the community-scale biochar system, a sensitivity analysis was conducted by varying five critical parameters: (1) WCO price (±20 %), (2) electricity tariff (±10 %), (3) biochar selling price (±15 %), (4) labor cost (±10 %), and (5) biochar yield (±10 %). The variation ranges were not arbitrarily assigned but rather reflect realistic fluctuations based on field observations, regulatory constraints, and market dynamics in rural Indonesia. For example, WCO cost is highly volatile due to its informal sourcing, justifying a broader ±20 % range. In contrast, electricity tariffs and labor wages are more stable, so they are assigned a ±10 % variation. The biochar selling price reflects moderate uncertainty due to market competition and seasonal demand (±15 %), while biochar yield may vary depending on biomass quality and process conditions (±10 %).

The effect of each variable change on profitability was quantified using the following formulation:

$$\Delta\pi = \pi_{adjusted} - \pi_{baseline} \tag{4}$$

where $\pi_{baseline}$ refers to the net profit per batch under default conditions (IDR 3500/kg for WCO, IDR 1500/kWh for electricity, IDR 300,000/batch labor cost, 60.95 % biochar yield, and IDR 12,500/kg selling price). $\pi_{adjusted}$ reflects profit under each altered scenario.

3.5. Investment, scaling strategy, and environmental considerations

To ensure broader replicability and strategic deployment of community-scale biochar systems, this study outlines a tiered investment framework integrated with potential policy support mechanisms. The model envisions a scalable structure, beginning with small-scale units and progressing to regional production hubs (Ulloa-Murillo et al., 2022). Tier 1 represents batch-based systems with a 300 kg capacity, requiring an investment of IDR 150 million—suitable for farmer groups or youth cooperatives. Tier 2 targets cooperative clusters with a capacity of 600–900 kg and an investment of IDR 300–450 million, while Tier 3 involves BUMDes-managed multi-batch or continuous systems (1–2 t/day) with a CAPEX ranging from IDR 600 million to 1 billion. Table 4 summarizes this scaling model.

Policy scenarios simulated include electricity subsidies (reducing tariffs from IDR 1500 to 1000/kWh), soft loans (e.g., *Kredit Usaha Rakyat*/KUR), and inclusion in carbon offset markets with potential revenue from CO_2 sequestration valued at USD 10–20/t. These mechanisms significantly improve project viability and resilience. The monetization of these carbon offsets through voluntary markets can further enhance the financial appeal of community-scale biochar initiatives, especially in tropical developing regions where sustainable livelihoods and climate resilience are priorities.

Beyond economics, environmental co-benefits are integral to the justification of biochar systems. Controlled pyrolysis avoids methane and particulate emissions from open burning, stabilizes carbon in the soil, and improves land productivity. Based on emission factors reported in recent life cycle assessments (Sahoo et al., 2021), the conversion of 1 t of OPF into biochar may offset up to 1.8 t-CO_2-eq. Embedding biochar within sustainable agriculture, waste valorization, and climate mitigation frameworks amplifies its role in circular bioeconomy strategies for rural areas.

The premium-grade biochar produced via optimized co-torrefaction—rich in fixed carbon (>70 %), low in ash (<5 %), and finely powdered—is primarily sold for horticulture, organic farming, and urban soil improvement due to its ability to enhance porosity, moisture retention, and nutrient-holding capacity. Its carbon-rich profile also

Table 1
Summary of prior techno-economic assessments of biochar systems.

Study	Feedstock	Process	Capacity	IRR (%)	PP (Years)	Remarks
Hu et al. (2024)	Swine manure	Pyrolysis	8000 t/y	11	4.6	Large-scale piggery
Zepeda et al. (2024)	Rice straw	Pyrolysis	10 t/h	14	14	Produces biochar-based slurry fuel
Parthasarathy et al. (2024)	OPF	Pyrolysis	20 t/h	22	4.6	No pretreatment
This study	OPF + WCO	Acid-pretreat + Co-torrefaction	300 kg/batch	>29	2.1	Community-scale, Southeast Asia



Fig. 2. Process flow of acid pretreatment and co-torrefaction of OPF with WCO in a 300 kg/batch system, producing premium-grade biochar for soil and carbon applications.

Table 2
Estimated capital investment breakdown for a 300-kg/batch biochar production system.

Component	Estimated cost (IDR)	Description
Equipment purchase	60,000,000	Reactor, dryer, mixer, filter, etc.
Equipment installation	18,000,000	Piping, cabling, and base construction
Instrumentation & controls	7,000,000	Thermocouples, valves, and safety systems
Civil works (building/shed)	15,000,000	Shelter for unit operation
Engineering & contingency	20,000,000	Design, unforeseen costs (~20 %)
Working capital (3 months)	10,000,000	Labor, electricity, and materials for the startup
Total estimated investment	**130,000,000**	

Note: Total cost is rounded to Indonesian Rupiah (IDR) 150,000,000 (~USD 9600) to account for variability in rural procurement, inflation margin, and exchange rate fluctuations.

Table 3
Operational cost components for 300-kg batch biochar production.

Cost component	Quantity per batch	Unit cost (IDR)	Total cost (IDR)	Notes
OPF	300 kg	800/kg	240,000	Collection, handling, and drying
WCO	300 kg	3500/kg	1,050,000	Sourced from local vendors
Electricity (inc. pretreatment)	68 kWh	1500/kWh	102,000	Heating, mixing operations, and thermal pretreatment
Labor	1 operator	–	300,000	Per batch payment
Consumables	Lump sum	–	90,000	Acid, water, PPE, etc.
Total operational cost	–	–	**1,782,000**	

Note: All values are presented in IDR. The total cost is equivalent to approximately USD 114, using the exchange rate of 1 USD = IDR 15,610.

Table 4
Tiered investment scaling model for biochar production.

Tier	Description	Capacity	Estimated CAPEX (IDR)	Institutional form
1	Community-scale batch system	300 kg/batch	~150 million	Farmer group/youth startup
2	Cooperative cluster unit	600–900 kg/batch	300–450 million	Multi-village cooperative
3	BUMDes-managed multi-batch system	1–2 t/d	600 million–1 billion	BUMDes

makes it a potential substitute for coal in low-emission energy systems (Lehmann and Joseph, 2024; Meyer et al., 2011). Growing interest in carbon sequestration has further driven demand in voluntary carbon markets. A 2024 field survey of five commercial retailers in Java and Sumatra reported a monthly demand of 3–5 t, with retail prices ranging from IDR 12,500–15,000/kg for small packaged products—highlighting its economic feasibility for decentralized rural production.

4. Results

4.1. Biochar yield and technical properties

The co-torrefaction of acid-pretreated OPF and WCO at 250 °C resulted in a solid biochar yield of 60.95 % and an energy yield of 87.43 % based on dry feedstock input. The HHV of the resulting biochar reached 31.97 MJ/kg, surpassing typical torrefied biomass values and positioning it within the premium-grade range for biochar fuels. These values were adopted from experimental data by Mutamima et al. (2025) and used as input parameters in this study's mass-energy balance and financial modelling.

Physicochemical characterisation of the biochar confirmed a fixed carbon content exceeding 70 %, ash content below 5 %, and a particle size distribution under 2 mm. These properties were verified using proximate and ultimate analyses, including CHN analysis and thermogravimetric methods (LECO TGA701), as well as FTIR and XRD to assess carbon bonding and crystalline features. A comparative summary of key performance metrics from related studies is provided in Table 5.

In sum, the co-torrefaction of OPF and WCO produces a technically robust biochar that is competitive across key performance indicators, particularly in terms of energy yield and product quality. These properties justify its selection as the core product for community-scale economic modeling, while its multipurpose functionality makes it especially suited for integrated rural development applications in biomass-rich tropical regions.

4.2. Operational cost results

The operational cost per 300 kg batch was IDR 1,782,000 (~USD 114), comprising OPF (IDR 240,000), WCO (IDR 1,050,000), electricity (IDR 102,000), labor (IDR 300,000), and consumables (IDR 90,000). These costs reflect local market conditions and field-validated rates. Based on a yield of 60.95 %, the system produced 183 kg of biochar per batch, resulting in a unit cost of IDR 9738/kg (approximately USD 0.62/kg). A detailed breakdown is provided in Table 3.

4.3. Financial viability and market scenarios

Financial projections under three selling price scenarios—IDR 12,500 (conservative), IDR 15,000 (moderate), and IDR 17,500/kg (optimistic)—showed strong profitability across all conditions. With a production cost of IDR 9738/kg and batch output of 183 kg, net profit per batch ranged from IDR 505,500 to 1,420,500. The corresponding PPs varied from 3.1 years to under 9 months, and BEP ranged from 297 to 106 batches. The IRR exceeded 29 % in all scenarios, reaching over 80 % under the optimistic case. A summary of key indicators is presented in Table 6.

4.4. Sensitivity analysis results

Sensitivity analysis was conducted by varying five key parameters: WCO price (±20 %), electricity tariff (±10 %), labor cost (±10 %), biochar yield (±10 %), and selling price (±15 %). Among these, selling price and biochar yield had the most significant effects on profitability.

Table 5
Comparison of biochar properties from OPF and similar biomass feedstocks.

Study	Feedstock	Treatment method	Temp. (°C)	Mass yield (%)	HHV (MJ/kg)	Energy yield (%)
Mutamima et al. (2025)	OPF + WCO	Acid pretreatment + Co-torrefaction	250	60.95	31.97	87.43
Thaim et al. (2019)	OPF	Torrefaction only	260	86.30	17.81	93.50
Lam et al. (2019)	EFB + WCO	Microwave co-torrefaction	250	85.50	28.00	100
Parthasarathy et al. (2024)	OPF	Pyrolysis	400	25.00	18.00	–
Ahmed et al. (2020)	Sawdust	Pyrolysis	400	51.85	25.01	–
Park et al. (2014)	Rice straw	Slow pyrolysis	300	50.20	16.60	70

Table 6
Profitability under tiered biochar market prices.

Scenario	Selling price (IDR/kg)	Revenue/ batch (IDR)	Profit/ batch (IDR)	BEP (batches)	PP[a] (years)
Conservative	12,500	2,287,500	505,500	297	2.1–3.1
Moderate	15,000	2,745,000	963,000	156	1.3–1.9
Optimistic	17,500	3,202,500	1,420,500	106	0.7–1.1

[a] Assuming 2–3 batches/week, 48 weeks/year.

A 20 % increase in WCO price reduced the batch profit by IDR 120,000, while a 15 % decrease in the selling price (to IDR 10,625/kg) reduced profit by IDR 334,125. In contrast, a 10 % increase in biochar yield resulted in a profit of IDR 189,091, while a 10 % decrease led to a loss of IDR 262,875. Variations in electricity and labor costs had smaller impacts, ranging from IDR ±10,200 and ± 30,000, respectively. These findings indicate that securing a stable feedstock supply and optimizing biochar conversion efficiency are critical to economic viability. Key outcomes are summarized in Fig. 3.

4.5. Scaling results and carbon offset potential

A three-tier investment model was evaluated to assess scalability. Tier 1 (community scale) involves a 300 kg/batch system with an estimated CAPEX of IDR 150 million. Tier 2 (cooperative cluster) scales to 600–900 kg/batch with an IDR 300–450 million investments. Tier 3 (BUMDes-level) targets 1–2 t/day with capital needs of IDR 600 million–1 billion. Estimated investment and institutional forms are summarized in Table 4.

The potential for carbon offset was also assessed. Based on LCA-based emission factors, converting 1 t of OPF into biochar may offset up to 1.8 t-CO_2-eq (Sahoo et al., 2021). Under current production yields, each 300 kg batch could offset ~0.55 t-CO_2-eq. This offers additional monetization potential through voluntary carbon markets, where credits are valued at USD 10–20/t-CO_2-eq.

5. Discussion

5.1. Interpretation of biochar properties

The biochar derived from co-torrefaction of acid-pretreated OPF with WCO demonstrated a high mass yield of 60.95 % and an energy yield of 87.43 %, with an HHV of 31.97 MJ/kg (see Section 4.1). These values exceed the typical ranges for conventional torrefied biomass (usually <55 % mass yield and <30 MJ/kg HHV), suggesting that the presence of lipid compounds from WCO plays a crucial role in enhancing the fuel properties of the resulting biochar.

The elevated HHV and low ash content (<5 %) indicate a high concentration of fixed carbon, which aligns with previous findings that lipid-rich co-reactants can act as hydrogen donors and facilitate secondary reactions that stabilize the char matrix (Parthasarathy et al., 2024; Lam et al., 2019). Moreover, the relatively low torrefaction temperature (250 °C) used in this process typically yields low-grade biochar

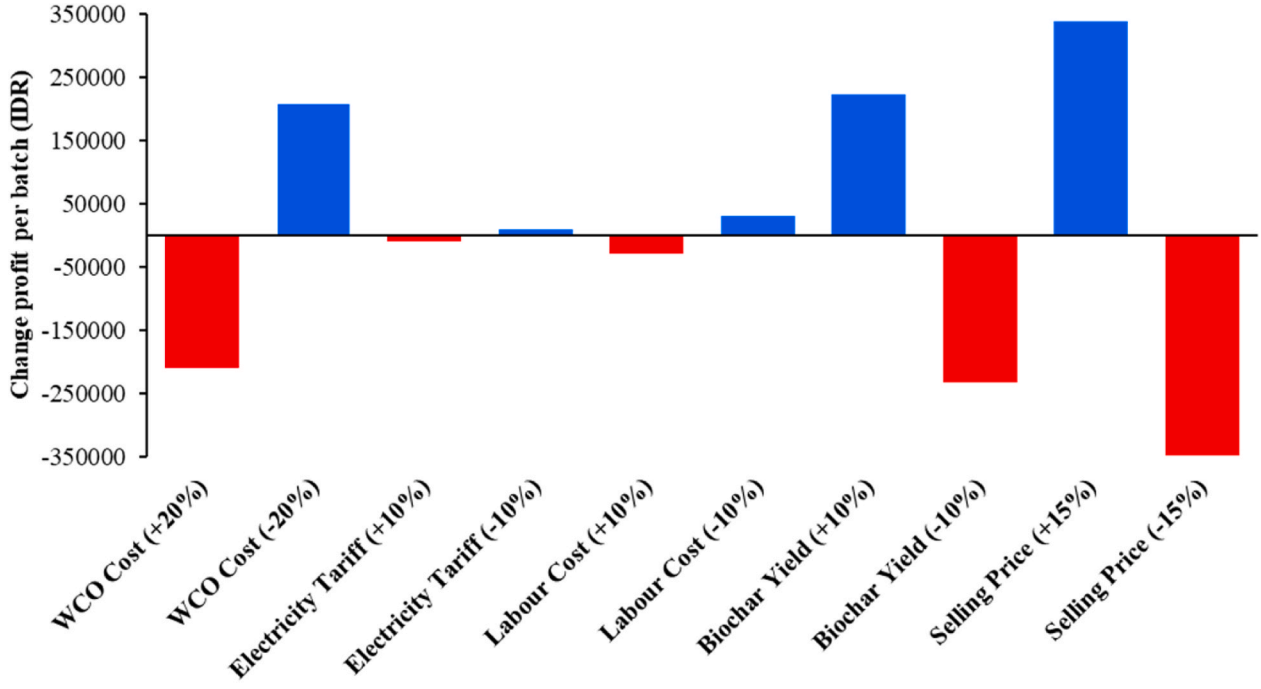


Fig. 3. Impact of parameter variations on profit per batch (IDR). Positive changes in blue; negative in red. Variables include WCO cost, electricity tariff, selling price, labor, and yield. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

with limited energy performance. However, the synergistic effect of WCO during thermal decomposition appears to have modified the devolatilization pathway, leading to superior carbon retention and thermal stability.

These observations are supported by the physicochemical quality of the biochar product described in Section 3.2, which showed particle sizes <2 mm and characteristics compatible with horticultural and carbon sequestration standards (Lehmann and Joseph, 2024). The integration of acid pretreatment prior to torrefaction likely contributed to the removal of alkali metals and hemicellulose, further enhancing fuel quality and mitigating the potential for slagging during use.

From a decentralized system perspective, the ability to produce high-quality biochar at relatively mild temperatures is advantageous for rural contexts where equipment durability and energy input are constrained. The use of locally available residues (OPF and WCO), coupled with a simple process configuration, allows for the development of low-cost, high-value biochar systems suitable for community-scale carbon management and soil amendment programs.

5.2. Feasibility of cost structure

The cost structure of the decentralized biochar system was purposefully designed to reflect rural operational conditions, favoring low-cost materials, simple equipment, and labor-intensive processing. As shown in Table 2, the estimated capital investment for a 300-kg/batch facility was approximately IDR 150 million (USD 9600), inclusive of reactor construction, auxiliary units (dryers, mixers, and filters), civil works, instrumentation, and working capital. This total reflects a Lang factor of 3.2, which is within the typical range for semi-automated, small-scale biomass processing systems (Turton et al., 2018). Such a factor indicates a cost-efficient design with a high potential for replication in village-level or cooperative settings.

The operational costs, summarized in Table 3, amount to IDR 1,782,000 (USD 114) per batch, resulting in a unit cost of IDR 9738/kg (USD 0.62/kg) for biochar. The cost composition is dominated by WCO procurement (IDR 1,050,000), followed by labor (IDR 300,000), and OPF (IDR 240,000). These figures reflect prevailing local market rates in Riau Province, Indonesia, and are applicable to other oil palm-producing regions with similar economic and agro-industrial characteristics (Owsianiak et al., 2021).

Crucially, the affordability of this system positions it as an inclusive innovation for rural entrepreneur networks, particularly youth and women's cooperatives. In contrast to centralized torrefaction or pyrolysis units—often requiring capital investment exceeding USD 100,000 and complex logistics (Hu et al., 2024)—this model offers an accessible entry point into the biochar value chain. The modular nature of the process—comprising chopping, acid pretreatment, blending, and torrefaction at 250°C—ensures low technical entry barriers, minimal automation requirements, and flexible workforce needs.

From a circular economy perspective, the valorization of OPF, an abundant agricultural residue, and WCO, a frequently mismanaged urban waste stream, exemplifies resource efficiency. The system effectively converts environmental burdens into economic value, supporting goals outlined in the ASEAN Framework for a Circular Economy for the ASEAN Economic Community (ASEAN Secretariat, 2021). Moreover, the initiative aligns with Indonesia's Green Economy Index (Bappenas, 2022), which emphasizes community-scale green business and local resource use.

The system's design also facilitates synergies with other village-based programs, including agroforestry, sustainable land management, and community composting. Such integration enhances social legitimacy and broadens participation, enabling economic, environmental, and social co-benefits. The structured yet flexible cost framework underscores the economic feasibility of the proposed model. Its replicability, low energy intensity, and compatibility with informal rural supply chains provide a resilient foundation for broader dissemination—without reliance on heavy subsidies or advanced infrastructure, which often constrain scalability in developing country contexts (Owsianiak et al., 2021).

5.3. Profitability and economic resilience

The decentralized biochar production model evaluated in this study demonstrates strong financial viability across a range of realistic market conditions. With a production cost of IDR 1,782,000 per batch (~USD 114), and a yield of 183 kg of biochar per 300-kg input, the unit production cost is approximately IDR 9738/kg (~USD 0.62/kg). At a conservative market price of IDR 12,500/kg, the model generates a batch revenue of IDR 2,287,500, resulting in a net profit of IDR 505,500 (~USD 32). In the moderate scenario (IDR 15,000/kg), profit increases to IDR 963,000, while the optimistic scenario (IDR 17,500/kg) yields IDR 1,420,500 in profit per batch. These correspond to profit margins between approximately 22 % and 44 %, reflecting robust economic performance even in conservative market settings.

Given a capital investment of IDR 150 million (~USD 9600), the BEP is estimated to be between 106 and 297 batches, depending on the market scenario. This translates to a PP of 0.7–3.1 years, assuming a moderate operational rate of 2–3 batches per week over 48 weeks annually. Such a PP is considered favorable for smallholder cooperatives and rural enterprises, and aligns with decentralized biomass systems previously reported by Nematian et al. (2021) and Owsianiak et al. (2021). As shown in Fig. 3, the sensitivity analysis highlights that the selling prices of biochar and WCO are the most influential variables. A 15 % increase in biochar price raises batch profit by over IDR 330,000, while a 20 % increase in WCO cost reduces profit by about IDR 120,000. Variations in labor and electricity costs were relatively minor (<10 %), underscoring the system's efficiency and operational simplicity.

The IRR is estimated at 29.2 % under conservative conditions, increasing to 71.9 % in moderate and 110.9 % in optimistic pricing scenarios. These values demonstrate the investment appeal of decentralized biochar systems and confirm their financial robustness under varied economic conditions (Sahoo et al., 2021; Owsianiak et al., 2021). Economic resilience can be enhanced by diversifying WCO sources—e.g., integrating institutional kitchens or tapping into industrial by-products like palm fatty acid distillate (PFAD)—which helps stabilize input costs (Beghetto, 2025). Additionally, incorporating solar-assisted drying or waste heat recovery can reduce energy costs and improve environmental performance (Amjed et al., 2024).

Beyond market sales, each batch also sequesters approximately 0.55 t-CO_2-equivalent. At voluntary carbon prices of USD 10–20/t, this provides an additional income of USD 5–11 per batch, potentially increasing profit margins by up to 15 %, and supporting eligibility for carbon offset schemes (Lehmann and Joseph, 2024; Vitelli, 2024).

5.4. Risk analysis and market barriers

Although the techno-economic evaluation supports the feasibility and profitability of decentralized biochar production using OPF and WCO, several risks could impede long-term implementation and replication. These include uncertainties in feedstock supply, labor availability, climatic conditions, market demand, and regulatory clarity—especially in rural areas where institutional capacity is often limited.

A central operational risk is the volatility of WCO availability and pricing, which exerts a disproportionate influence on profitability. As highlighted in the sensitivity analysis (Fig. 3), a 20 % increase in WCO cost can lower batch profit by approximately IDR 210,000 (~USD 13.50), significantly reducing the financial attractiveness in areas with sparse food service establishments or fragmented informal WCO markets. This finding is consistent with those of Dace et al. (2024) and Lokesh et al. (2018), who observed that inconsistent supply chains are a significant barrier to circular bioeconomy models involving used oils. To

mitigate this risk, cooperatives or local enterprises could establish fixed-price contracts, pooling arrangements, or barter mechanisms with food vendors. Additionally, alternative lipid sources, such as palm fatty acid distillate (PFAD) or expired oils from canteens and food processors, could supplement WCO inputs (Beghetto, 2025).

Labor is another constraint, particularly in areas facing seasonal outmigration or ageing rural populations. Despite the system's labor-lean design, which requires only one trained operator per batch, essential steps such as acid handling, process control, and reactor safety still demand basic technical competence. Similar issues have been observed in small-scale biomass systems across Southeast Asia (Sovacool and Drupady, 2016). Embedding community-based technical training, with modules developed jointly by vocational schools, agricultural extension services, or NGOs, could ensure local capacity for safe and consistent operation.

Environmental conditions present additional challenges. In tropical climates like Riau, high humidity during rainy seasons can compromise OPF drying and storage, which in turn affects process efficiency and biochar quality. Without adequate protection, stored OPF may become moldy, while WCO may degrade due to microbial growth. These risks can be addressed through solar-assisted dryers or biomass-fueled drying systems, which are low-cost and compatible with decentralized operations (Amjed et al., 2024).

On the demand side, limited awareness and weak regulatory frameworks for biochar quality present significant barriers. In Indonesia, there is no nationally enforced biochar certification or grading system, leading to market opacity and difficulty in distinguishing high-quality biochar from inferior products. This hinders value capture and consumer trust, especially in sectors beyond informal horticulture (Goenadi & Santi, 2017). Establishing voluntary standards, supported by local governments, farmer groups, or agricultural NGOs, could serve as a transitional step while national policy evolves.

Finally, institutional and legal uncertainties constrain broader adoption. The classification of WCO—whether as waste, fuel, or feedstock—is often unclear, resulting in conflicting bureaucratic requirements across ministries (e.g., energy, agriculture, environment). This can delay permits, block access to microcredit schemes such as *Kredit Usaha Rakyat (KUR)*, and reduce investor confidence. Aligning policy instruments through integrated waste management guidelines and recognizing community-scale WCO utilization as part of the circular economy agenda would help reduce these regulatory frictions (Bappenas, 2022).

In summary, while the proposed system is economically sound and technically modular, its success depends on anticipating and managing contextual risks. Tailored mitigation strategies—ranging from technical training and contractual supply agreements to local certification schemes and enabling policy instruments—will be crucial to ensure sustainable and inclusive deployment.

5.5. Scaling strategies and institutional models

Scaling up the decentralized biochar production system necessitates not only technical and economic viability, but also institutional frameworks that support replication across diverse rural contexts. As illustrated in Fig. 4, higher production capacities—such as those managed by cooperatives or village-owned enterprises—are associated with improved NPV performance. Specifically, deploying five reactor units under coordinated management, each operating at 80 % capacity, could generate net annual profits exceeding IDR 100 million (~USD 6400), making the system viable for village-scale income generation.

The system's inherent modularity and process simplicity enable flexible adaptation to local contexts without requiring extensive engineering alterations. Similar modular models have been adopted in rural India and Sub-Saharan Africa for decentralized biomass valorization (Chaudhary et al., 2022; Mperejekumana et al., 2024). However, achieving meaningful scale requires institutional anchoring—particularly through cooperatives, youth farmer groups, or village-owned enterprises (BUMDes)—that can aggregate production, manage resources collectively, and reduce transaction costs.

Operationally, centralized preprocessing hubs—for WCO collection, OPF drying, or acid handling—could streamline workflows and improve consistency in feedstock quality. Labor pooling and task rotation mechanisms, especially in labor-scarce villages, can enhance workforce flexibility while building community skills (Azizi et al., 2010). Additionally, collaborations with local universities, NGOs, and agricultural training centers can facilitate quality assurance, capacity development, and innovation diffusion, including the use of digital tools for real-time process monitoring or carbon tracking (Causa et al., 2025).

One of the most pressing barriers to scaling is access to finance. Initial capital costs—though moderate—may be unaffordable for many rural actors without external support. National credit schemes such as *Kredit Usaha Rakyat (KUR)*, *Dana Desa*, or green financing platforms (e.g., Environmental Fund Management Agency – BPDLH in Indonesia) offer avenues for concessional loans or startup grants (Bappenas, 2022).

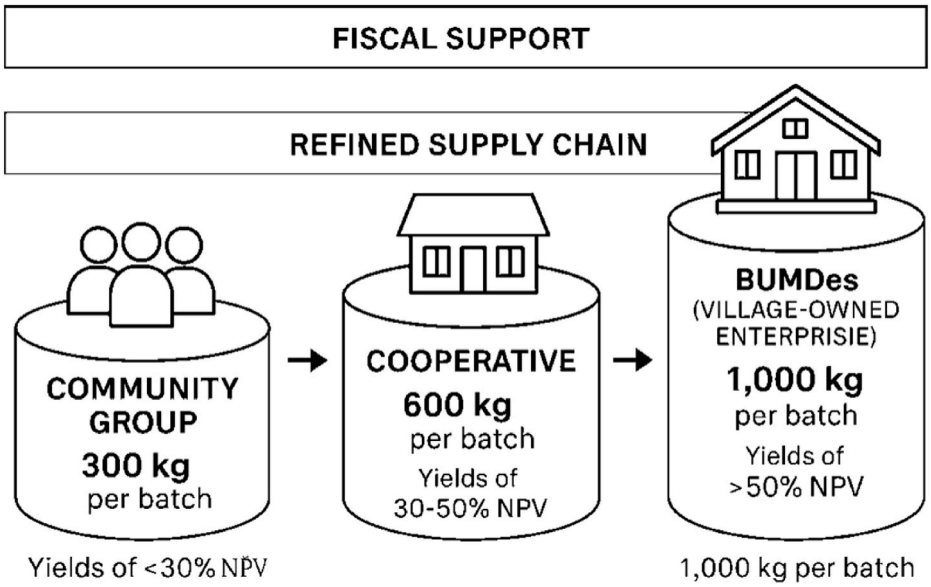


Fig. 4. Biochar investment scaling model.

In contexts where climate-smart agriculture is prioritized, results-based financing (RBF) or carbon-linked grants may incentivise broader adoption (Salma et al., 2024).

Institutionalisation is equally critical. Integrating biochar initiatives into local development plans (RKPDes) and securing recognition under village regulations can promote legitimacy, facilitate coordination with other programmes (e.g., waste management or reforestation), and provide a basis for community-level investment. Empowering women and youth, particularly in downstream activities such as packaging, distribution, or educational outreach, can expand the socio-economic impact of the initiative while promoting inclusivity.

In essence, the successful replication of this decentralized model hinges on a three-pillar integration: technical modularity, financial accessibility, and institutional embedding. Positioning biochar production within existing rural governance ecosystems—through cooperatives or BUMDes—not only improves economies of scale and operational resilience but also fosters a sense of community ownership, which is fundamental for long-term sustainability and scalability across rural Indonesia.

5.6. Policy instruments and carbon credit potential

The viability and scalability of decentralized biochar systems are significantly influenced by the policy landscape and incentive structures that govern rural enterprises, waste valorization, and climate action. In Indonesia, the current policy framework—though supportive in rhetoric—remains fragmented in execution, particularly in integrating low-tech innovations, such as biochar, into formal rural development and climate mitigation schemes.

At the national level, opportunities exist within the FOLU Net Sink 2030 strategy, which aims to achieve a net carbon sink from the forestry and land use sector by 2030 (Ministry of Environment and Forestry of Indonesia, 2022). Biochar, as a soil amendment with demonstrated carbon sequestration benefits, aligns well with this target. Moreover, the government's ongoing commitment to restore peatlands and degraded lands—including initiatives under BRGM and the Peat Restoration Agency—can serve as entry points for the deployment of biochar systems, especially given its capacity to improve soil structure, nutrient retention, and microbial resilience (Lehmann and Joseph, 2024).

Despite these alignments, biochar remains unregulated under most agricultural and environmental policy instruments. The absence of national standards on biochar quality, safety, and application protocols limits both trust and market penetration. This regulatory vacuum necessitates the interim adoption of international guidelines—such as those from the International Biochar Initiative (IBI) or the European Biochar Certificate (EBC)—while encouraging the development of localized certification mechanisms through universities or regional research centers.

Carbon finance represents a potentially transformative pathway for enhancing the economic attractiveness of biochar. Each kilogram of biochar can sequester approximately 2.03 kg of CO_2-equivalent, depending on the feedstock and process parameters (Murphy, 2025). When scaled across multiple community units, this sequestration potential becomes quantifiable and tradable. However, to participate in voluntary carbon markets, robust MRV (Monitoring, Reporting, and Verification) frameworks are required—something currently lacking at the grassroots level.

To bridge this gap, partnerships with academic institutions or blockchain-based MRV providers can be explored. Additionally, initiatives such as Indonesia's Nusantara Carbon Scheme (NCS) and international programs like Verra and the Gold Standard offer methodologies for quantifying the carbon benefits of biochar application. Early participation in these schemes would not only provide additional income streams to community producers but also enhance transparency and investor confidence.

On the financing side, inclusion of biochar systems under green credit lines—such as *Kredit Usaha Rakyat* (KUR Hijau), or the BPDLH—would enable broader access to startup capital. Likewise, integration into government procurement schemes for land rehabilitation or village development programmes could catalyse institutional demand. Finally, local governments can play a catalytic role by embedding biochar systems into the Regional Climate Change Action Plan (RCCAP), issuing regional regulations for bio-based fertilizers, or offering matching grants for equipment procurement. Such multilevel policy integration is critical to scale the technology while ensuring social equity, environmental integrity, and economic inclusivity.

6. Conclusion

This study provides a novel techno-economic assessment of community-scale biochar production using acid-pretreated OPF and WCO through co-torrefaction. The system achieved a solid yield of 60.95 %, an energy yield of 87.43 %, and a heating value of 31.97 MJ/kg, confirming its technical viability for decentralized biomass valorization. With a production cost of IDR 9738/kg (~USD 0.62), it remains financially feasible under moderate to optimistic pricing scenarios, delivering a PP below 1.1 years and an IRR exceeding 40 % in high-price contexts. Beyond technical and financial metrics, the study advances the literature by integrating process optimization with financial modelling and policy-relevant investment tiers—offering a replicable framework for rural deployment. The model supports carbon sequestration goals, enables smallholder income diversification, and aligns with circular economy initiatives in palm-oil-producing regions. However, the findings are based on secondary data and scenario modelling without field trials. Future studies should validate reactor performance under real conditions, monitor emissions, and explore other biomass feedstocks. The integrated approach—combining a comprehensive cost estimation framework, expanded operational and sensitivity analyses, environmental impact considerations, and policy-aligned scaling strategies—provides robust evidence for the technical, economic, and environmental feasibility of decentralized biochar systems in biomass-rich rural regions. Overall, the system presents a scalable, low-emission solution for sustainable rural development, with clear applicability for clean energy transitions and inclusive bioeconomy policies in Southeast Asia.

CRediT authorship contribution statement

Indra Purnama: Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization. **Anisa Mutamima:** Writing – original draft, Visualization, Investigation, Formal analysis, Data curation. **Sunarno Sunarno:** Validation, Methodology, Investigation. **Imran Al Ucok Nasution:** Validation, Software, Formal analysis. **Eka Syaputra:** Investigation, Formal analysis, Data curation. **Muhammad Aziz:** Writing – review & editing, Supervision, Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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